Fitting a curve to the dam – Results

NM-LAB-08252026 · Laboratory Activity 02 · Numerical Methods BES6-M · n=288 readings, 2026-07-21 00:00 to 2026-07-23 23:45, dt=0.25 h · Analysed by Jamison

Model and defense

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h(t) = c + a1 / (1 + exp(-k1(t - t01))) + a2 / (1 + exp(-k2(t - t02)))
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The record climbs slowly, surges through one flood pulse peaking near hour 35, then recedes to a lower plateau, so it never settles toward a single ceiling the way one logistic or Gompertz would. A sum of two logistics — one positive-amplitude term for the rise, one negative-amplitude term offset later for the recession — lets the same functional family model inflow exceeding outflow during the pulse and then falling back below it.

Fitted parameters

Parameter	Estimate	SE	t	p (two-tailed)
c	14.3493	0.0097	1483.08	<0.0001
a1	8.9159	0.7531	11.84	<0.0001
k1	0.5117	0.0213	24.05	<0.0001
t01	31.5198	0.0906	348.09	<0.0001
a2	-3.7212	0.7594	-4.90	<0.0001
k2	0.2831	0.0138	20.58	<0.0001
t02	37.5462	1.0575	35.50	<0.0001

Goodness of fit

SSE (m^2)	1.8017	**SST (m^2)**	2034.0779
R^2	0.99911	**s, std. error of estimate (m)**	0.0801
n, p, degrees of freedom	288, 7, 281		

Derivatives

Maximum dh/dt = 0.960 m/h at t=29.25 h (2026-07-22 05:15).

The raw d2h/dt2 crosses zero 208 times over 288 points — almost all of that is the logger's own centimetre rounding getting amplified by differencing twice, exactly the fragility the brief warns about, not 208 real events. Smoothing over a 2-hour window shows a single real swing: dh2 stays positive while the rate of rise is still speeding up, crosses to negative at t=29.75 h (right where dh/dt itself peaks — the rate stops accelerating), and bottoms out at t=35.00 h, its point of hardest deceleration as the level crests. Together that says the inflow arrived as one accelerating-then-decelerating pulse, not a steady tap.

Reading the residuals

The two-half averages land near zero (first-half mean +0.0010 m, second-half mean -0.0010 m), which would look like a clean fit on summary statistics alone, but the residuals are not patternless: they run same-sign for 54 points at a stretch (13+ hours), and there are 5 separate runs at least 5 hours long covering almost the entire record — negative for the first ~11.5 h, positive through ~12-25 h, swinging hard through the surge itself, positive again for ~43-56 h, then negative for the last ~12.5 h. That is a shape problem, not noise: the two-logistic curve is a touch too straight through the slow pre-event rise and the post-event plateau, on top of not quite matching the sharpest part of the surge. The largest residual is 0.221 m, about 22x the logger's 1 cm resolution, so R2=0.9991 is a real number but it is sitting on patterned, not random, residuals — report it as a good fit with a known, systematic shortfall rather than a perfect one.

Area under the fitted level

A = 1260.971 m·h (quad, abs. error estimate 5.07e-07). Trapezoid cross-check on the raw readings: 1260.996 m·h (gap -0.025 m·h, -0.002%). quad integrates the smooth fitted curve while trapezoid sums straight-line chords between the noisy raw points, so the two disagree by the amount the logger's centimetre-level rounding and 15-minute gaps miss versus what the continuous model interpolates through them.

This area is the time-integral of depth above the gauge datum over the 71.75-hour record, in metre-hours: it summarises how high and how long the reservoir stood, not how much water moved. Turning it into a volume would need the reservoir's stage-area curve, which is not part of this dataset — flood control should read it as a loading/duration indicator, not a volume.

Prepared for NM-LAB-08252026 · stage log by Engr. Mark Airol Escranda · analysed by Jamison. All values on this page were computed in lab02_jamison.py and are reproduced here unchanged from lab02_jamison_results.json.